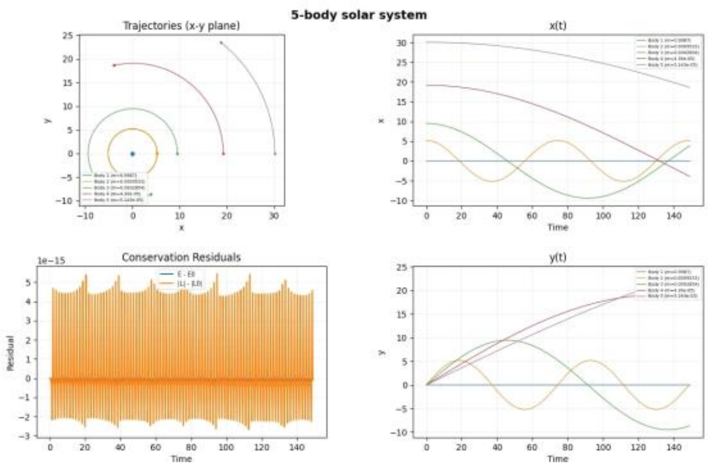
Problem 2

2a) I used the outer planets to verify 5BP - Sun, Jupiter, Saturn, Uranus, Neptune

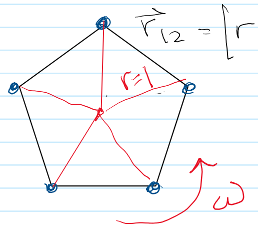
$m_{sun} = 1$ $m_J = 9.54 \times 10^{-4}$ $m_S = 2.86 \times 10^{-4}$ $m_U = 4.37 \times 10^{-5}$, $m_N = 5.15 \times 10^{-5}$

$V = m_{sun}/r$ $r_J = 5.2$ AU $r_S = 9.5$ AU $r_U = 19.2$ AU $r_N = 30.1$ AU

Energy & momentum are conserved thru 2 Jupiter orbits.



2b)



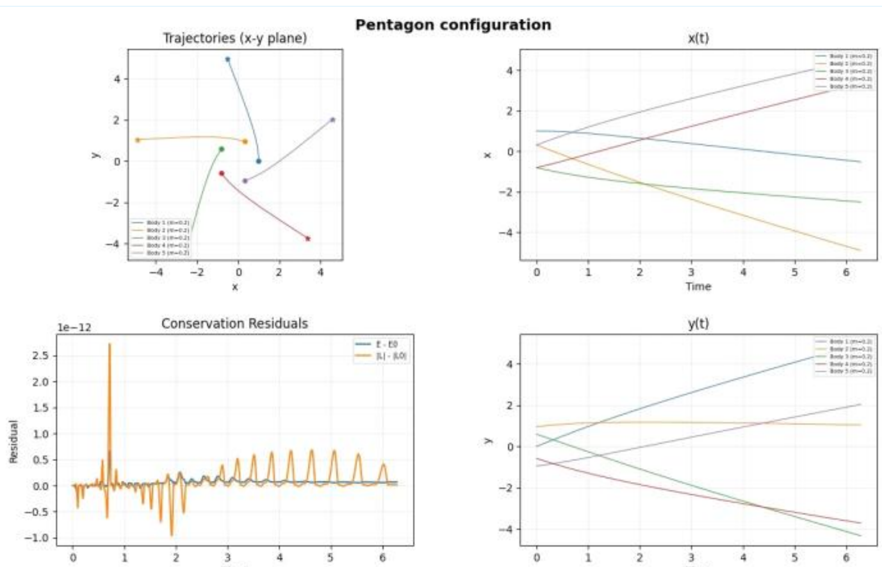
$\vec{r}_{12} = \left[r\cos\left(\frac{2\pi}{5}\right), r\sin\left(\frac{2\pi}{5}\right), 0 \right]$

$m_i = \frac{1}{5}$ $\omega = \frac{GM}{r^3}$

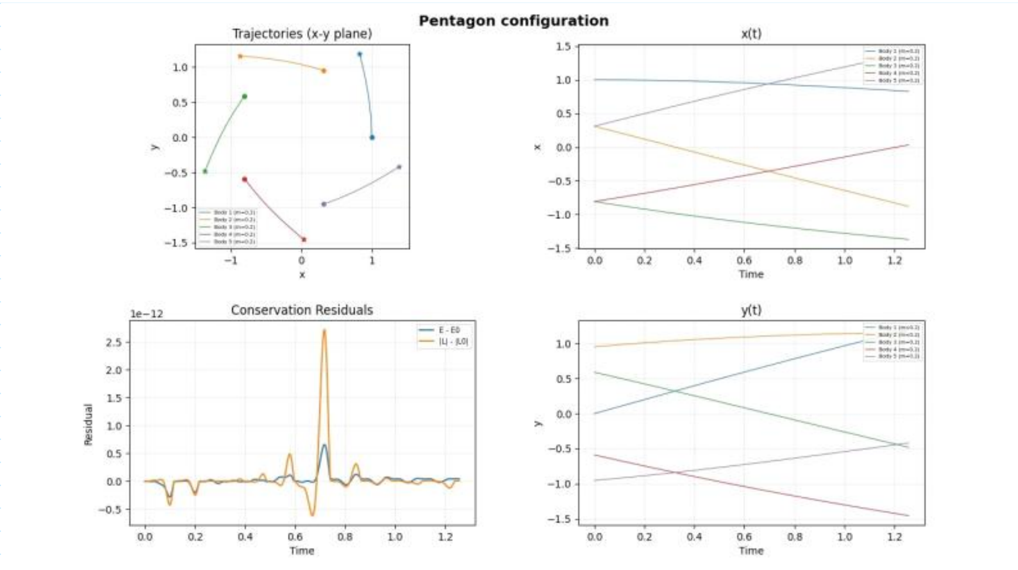
$v_i = \omega \vec{r}_i$

In this configuration the bodies fly away immediately.

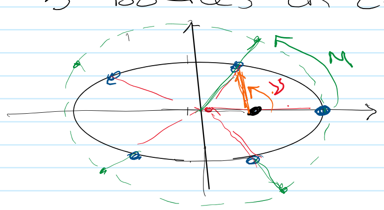
Here is one period:



Here is 0.2 period



2c) 5 bodies in eccentric orbit



$a = 1 \quad e = 0.1$

$h^2 = \mu a(1-e^2)$

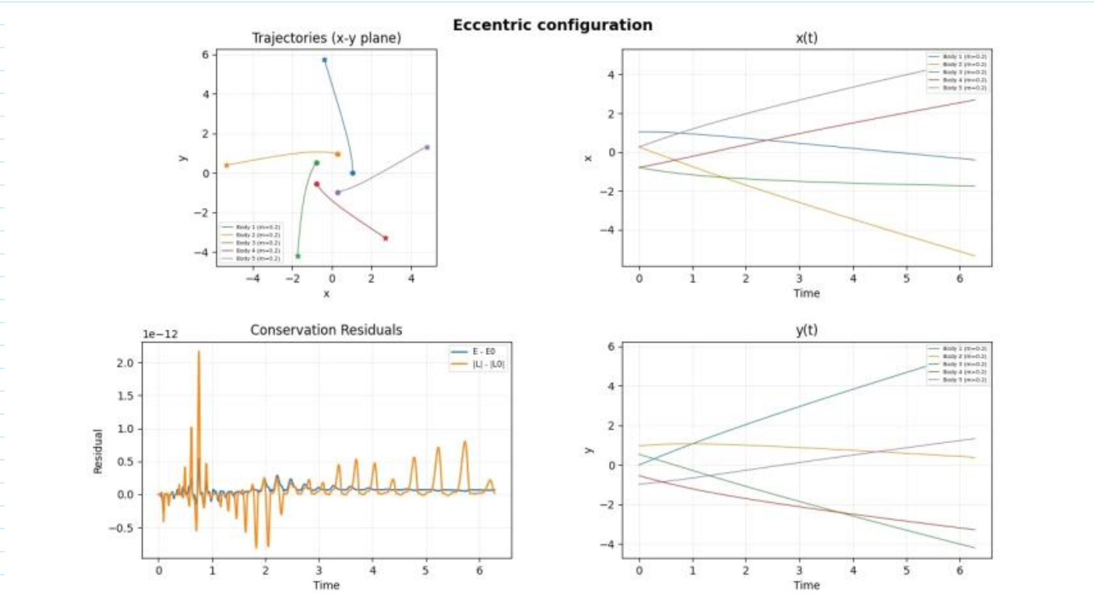
period $= 2\pi\sqrt{\frac{a^3}{\mu}}$

M = mean anomaly

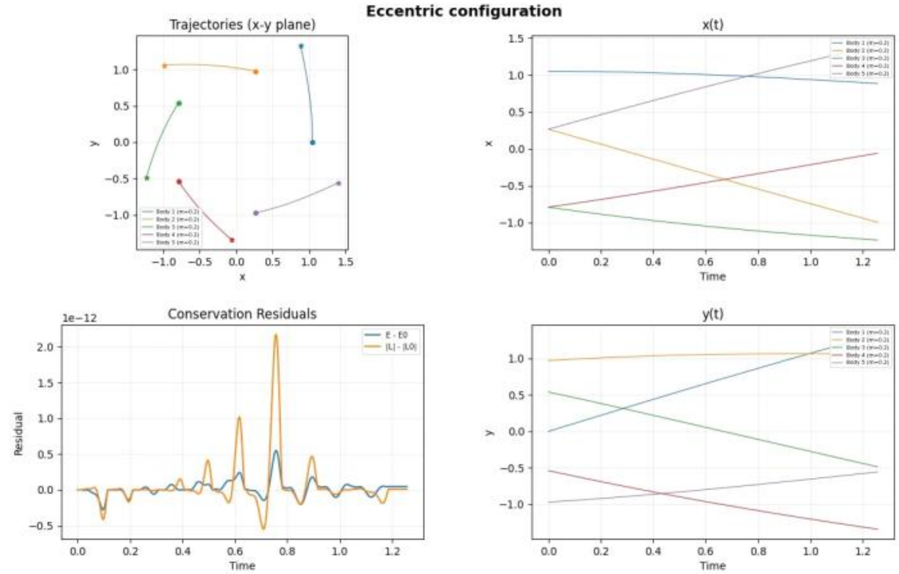
$f = E - E\sin(e) - M$

$\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2}$

The system is still unstable & the masses fly away from each other immediately but since there the masses have non uniform velocity the orbit departures are "shallower" since they are on an ellipse the velocity components are slightly different.

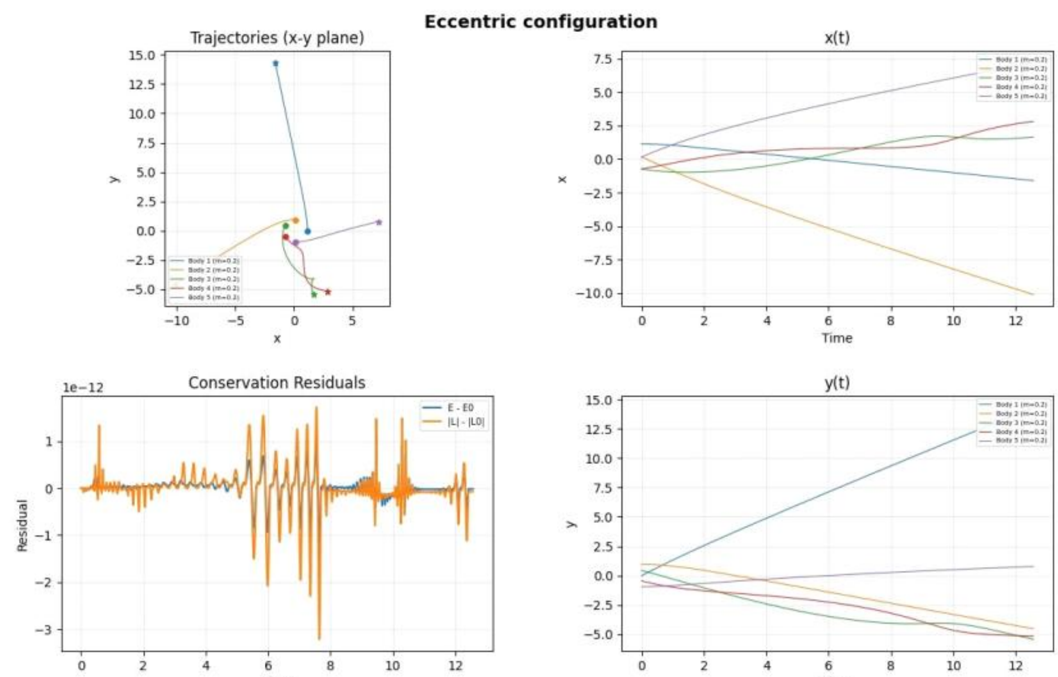


This 0.2 period

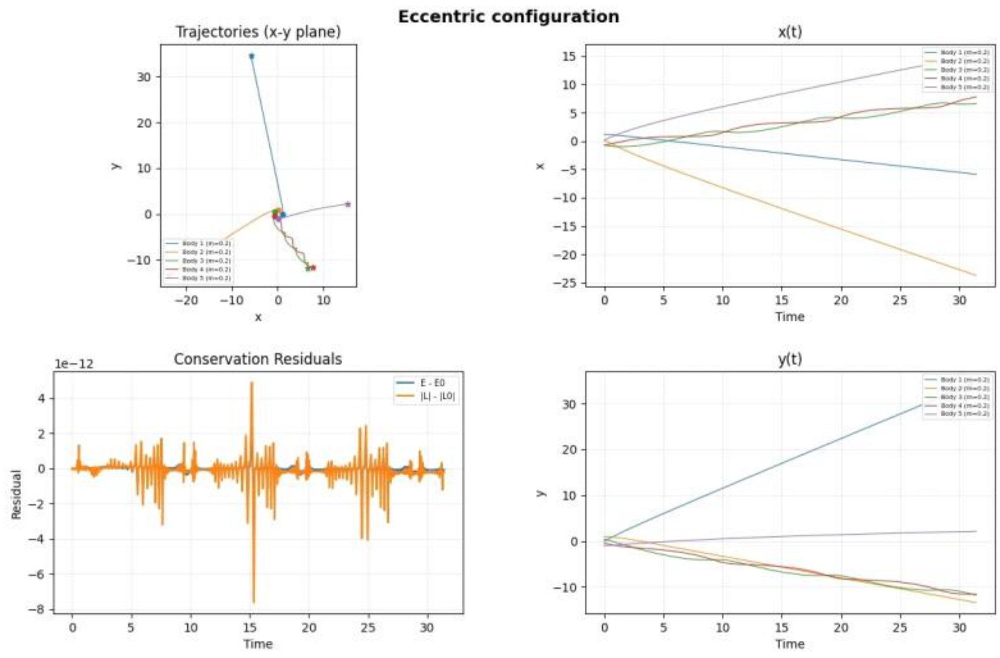


As the eccentricity is increased, the body @ periapsis leaves the system fastest & travels farther away than the other bodies.

This is e = 0.3 over the course of 2 periods. The bodies near apoapsis continue to impact the other's orbit because their velocity is slowest.

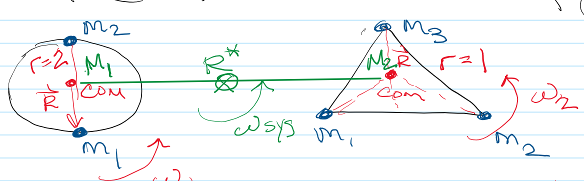


After 5 periods they are still bound to each other.



2d) 2 body & 3 body system
orbit each other & then fly by

$V = \sqrt{\frac{M_1+M_2}{r_{12}}}$



$\vec{R} = \frac{1}{\sqrt{3}}$

$r_{12} = 2$

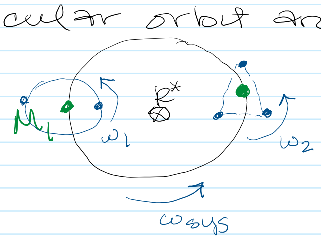
$V = \sqrt{\frac{2}{5/2}} = \sqrt{\frac{1}{5}}$

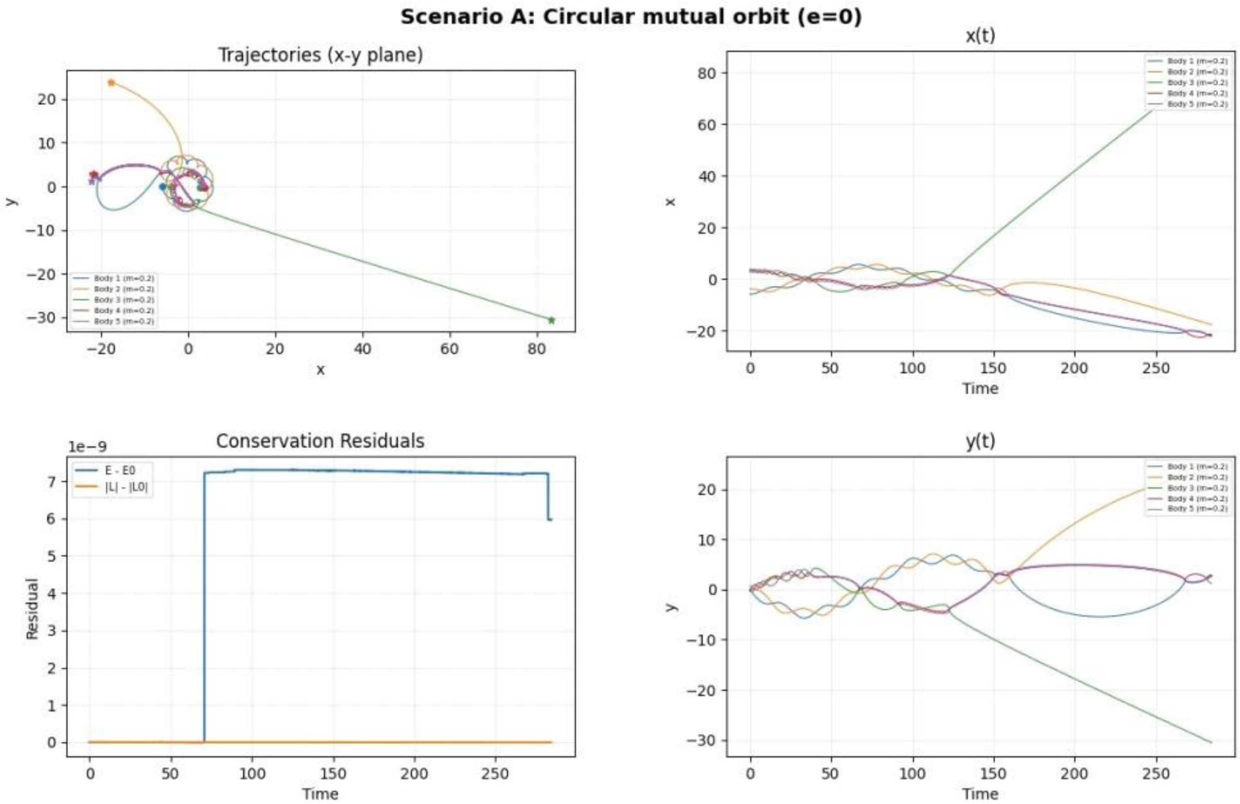
$r_{12} = r_{23} = r_{13} = 1$

$V = \sqrt{\frac{3}{5}}$

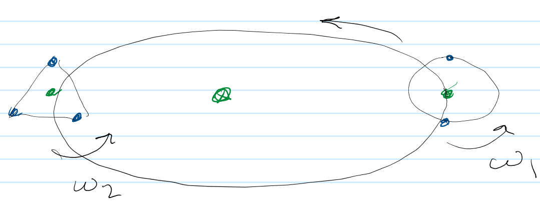
start w/ all masses equal: $m = 1/5$

A) circular orbit around mutual COM

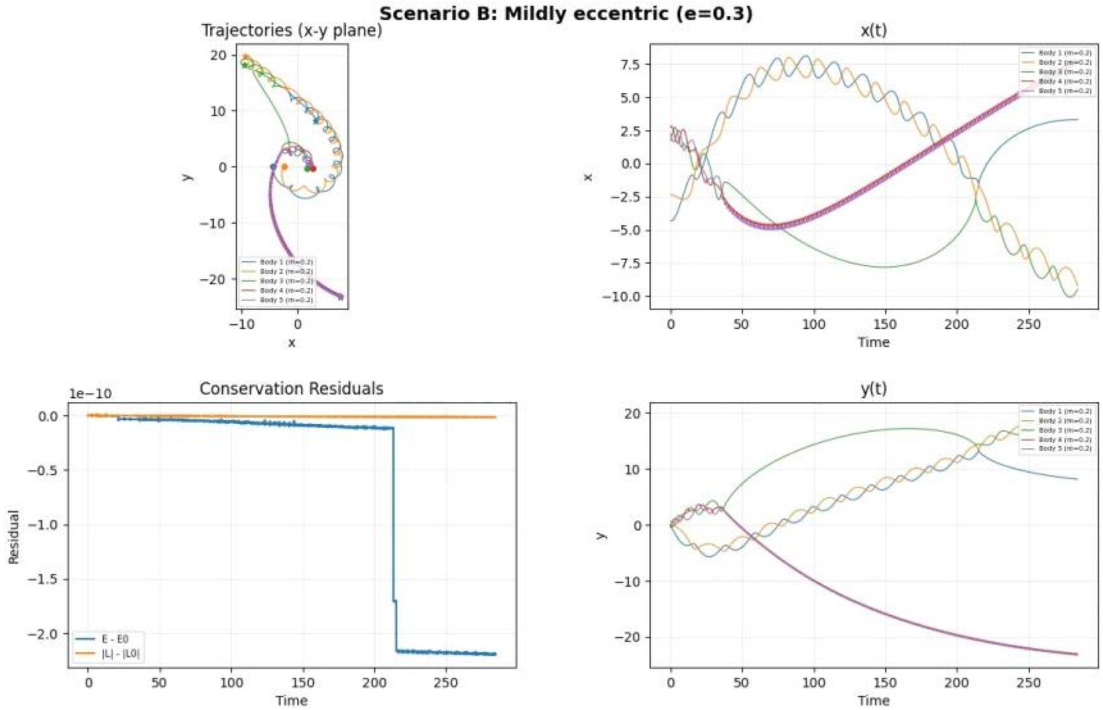




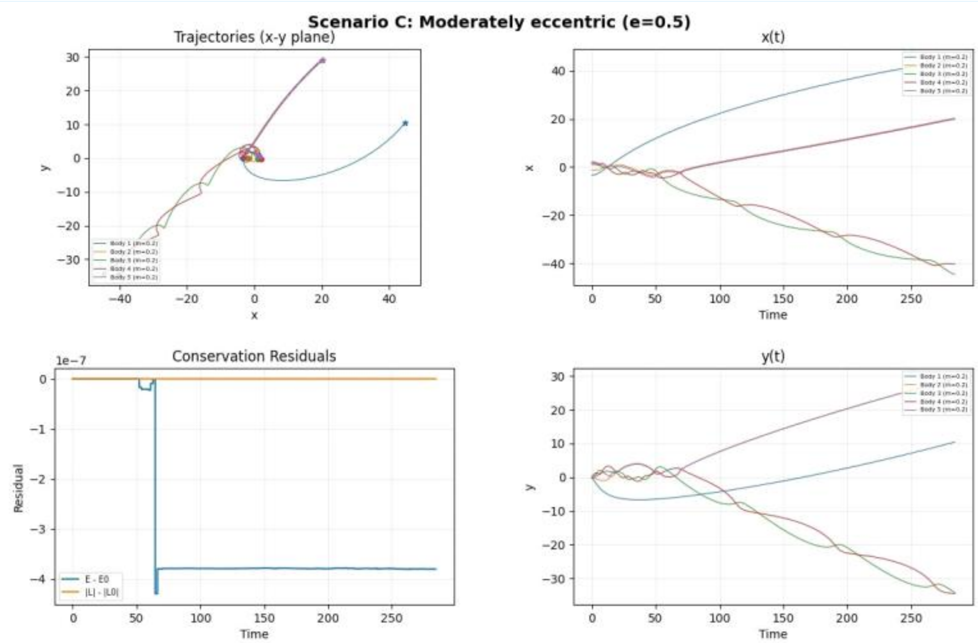
B) elliptical orbit around mutual Com



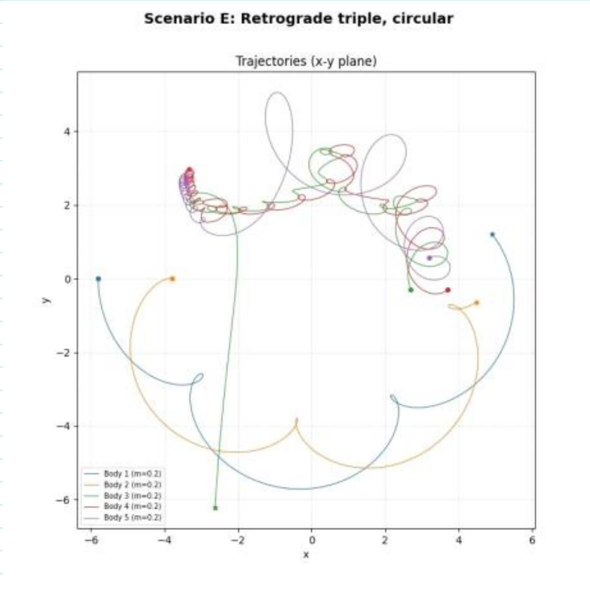
$\boxed{e = 0.3}$

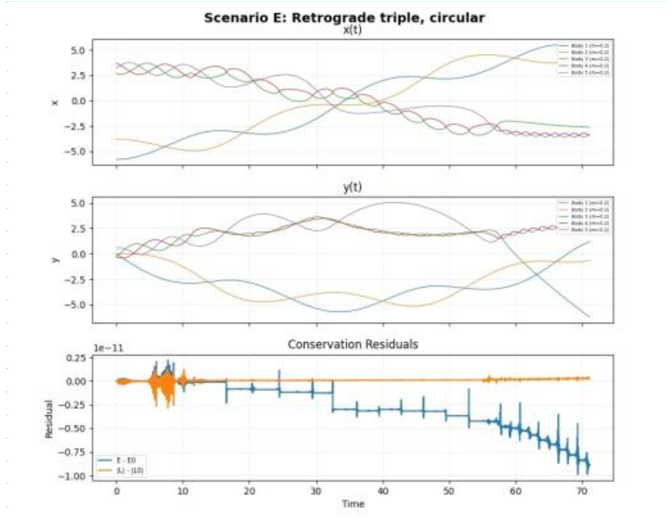


$\boxed{e = 0.5}$

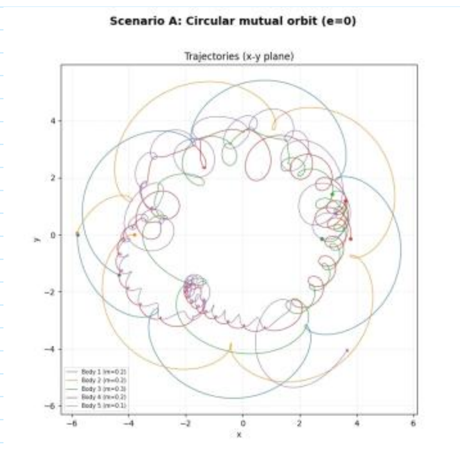


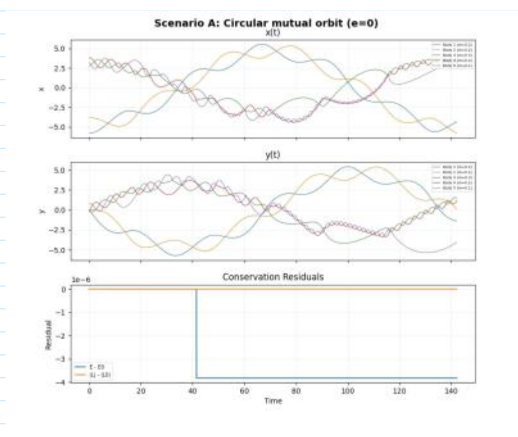
This has 3 body system rotating retrograde to 2 body system, in circular orbit





This example shows impact of non-equal masses & their impact on the trajectories of other bodies.





In all the scenarios I tested, given enough time at least one body was ejected. The motion is very chaotic & hard to predict. Lighter bodies are more impacted in their orbits when perturbed by other systems.